

Bridge Day 1 INSTRUCTOR KEY

Visible Output and Stored State

Learning sequence: Predict - Trace - Explain - Test - Interpret

Instructor key with answers in student response locations

Name		UIN		Section	
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Concrete engineering situation

Your team is doing a first-day hallway cart check. A small wheeled cart is rolled along a taped 12-meter path. One teammate times the run with a stopwatch. Another teammate enters the values in a Python notebook so the team can decide what should go into the lab record.

The problem today is not advanced physics. The problem is evidence: what did the notebook store, what did it visibly show, and which value is safe to write in the record?

Engineering task	Build a reliable evidence trail for two hallway cart trials before a result is copied into a lab record.
Computational move	Separate measurement inputs, stored variables, printed output, notebook display, and later overwrites.
Python focus	assignment, arithmetic expression, variable overwrite, print , final-expression notebook display, state after each cell.
Evidence to keep	exact visible lines, displayed values, stored values after each cell, a corrected lab note, and one non-mutating check.

Day 1 evidence rules

1. Assignment stores or overwrites a value. Example: `speed_mps = distance_m / time_s`.
2. `print(...)` creates visible output but does not by itself change a variable.
3. In a notebook, a bare expression at the end of a cell can display a value. Displaying a value is not the same as storing it.
4. A later assignment can overwrite an earlier stored value. Your trace must say what is stored *after each cell*, not just what looked important on the screen.

1. Read the notebook as an evidence system

recognize

predict

Use the scenario above to identify what kind of evidence each line can create. Do not calculate everything yet.

```
# Cell 1: first run on the hallway path
trial_id = "cart-A"
distance_m = 12.0
time_s = 3.0
speed_mps = distance_m / time_s
print("trial", trial_id)
print("speed", speed_mps)

# Cell 2: second run, same path, new time
trial_id = "cart-B"
time_s = 2.4
speed_mps = distance_m / time_s
print("trial", trial_id)
speed_mps + 0.2

# Cell 3: keep the adjustment separate
adjusted_speed = speed_mps + 0.2
print("stored", speed_mps)
print("adjusted", adjusted_speed)

# Cell 4: decide to record the adjusted value
speed_mps = adjusted_speed
print("recorded", speed_mps)
```

Pts.	Prompt	My evidence
1	Which variable names store measurement inputs rather than calculated results?	Key: distance_m and time_s. trial_id is a run label.
1	Which line first calculates a speed from the measured distance and time?	Key: speed_mps = distance_m / time_s
1	Which line in Cell 2 displays a candidate adjusted value without storing that adjustment?	Key: speed_mps + 0.2; no assignment stores it.
1	Which line in Cell 4 overwrites the stored value of speed_mps?	Key: speed_mps = adjusted_speed

2. Trace stored state after each cell

trace

predict

Complete the state table. If a variable does not exist yet, write **not defined**. The first row has hints; the later rows require you to track overwrites.

Checkpoint	trial_id	time_s	speed_mps	adjusted_speed	Reason for any change
After Cell 1	cart-A	3.0	Key: 4.0	not defined	Key: Initial measured speed: 12.0 / 3.0.
After Cell 2	Key: cart-B	Key: 2.4	Key: 5.0	Key: not defined	Key: Trial/time overwritten; speed recalculated. Final expression displays 5.2 but does not store it.
After Cell 3	Key: cart-B	Key: 2.4	Key: 5.0	Key: 5.2	Key: Adjustment stored separately; speed_mps remains measured speed.
After Cell 4	Key: cart-B	Key: 2.4	Key: 5.2	Key: 5.2	Key: speed_mps = adjusted_speed overwrites the speed variable.

Pts.	Prompt	My response
2	At what checkpoint does <code>speed_mps</code> first become 5.0? Explain the measurement change that caused it.	Key: After Cell 2; <code>time_s</code> changes to 2.4, so $12.0 / 2.4 = 5.0$.
2	At what checkpoint does <code>speed_mps</code> first become 5.2? Explain what line caused the overwrite.	Key: After Cell 4; <code>speed_mps = adjusted_speed</code> .

3. Separate printed output from notebook display

predict

trace

Visible evidence can come from printed output or from a final notebook expression. Complete both columns. If a cell has no notebook display, write **none**.

Cell	Printed output lines from <code>print(...)</code>	Notebook display from final bare expression
Cell 1	Key: <code>trial cart-A; speed 4.0</code>	Key: <code>none</code>
Cell 2	Key: <code>trial cart-B</code>	Key: <code>5.2</code>
Cell 3	Key: <code>stored 5.0; adjusted 5.2</code>	Key: <code>none</code>
Cell 4	Key: <code>recorded 5.2</code>	Key: <code>none</code>

Common trap to check

A displayed value can look official because it appears on the screen. But a lab record should say whether the value was printed, displayed as a candidate calculation, or stored as the current value after an assignment.

Pts.	Prompt	My response
2	In Cell 2, why is the displayed 5.2 not yet the stored value of <code>speed_mps</code> ?	Key: It is a bare expression result. No assignment stores it, so <code>speed_mps</code> remains 5.0.
2	In Cell 3, why is it useful to print both <code>stored</code> and <code>adjusted</code> ?	Key: It separates the current measured value from the adjusted candidate before overwrite.

4. Debug the lab record**debug****explain****test**

A teammate writes this lab note after Cell 2:

Lab note to debug

Trial cart-B speed was recorded as 5.2 m/s because the notebook showed 5.2.

Pts.	Prompt	My response
2	What is accurate in the teammate's note?	Key: The notebook did show 5.2 after Cell 2.
2	What is inaccurate or incomplete in the teammate's note?	Key: After Cell 2, 5.2 was only displayed. It was not yet stored/recorded in <code>speed_mps</code> ; stored <code>speed_mps</code> was 5.0.
2	Rewrite the note so it separates measured speed, displayed candidate adjustment, and the value actually recorded after Cell 4.	Key: After Cell 2, cart-B measured speed was stored as 5.0 m/s and 5.2 was only displayed as an adjustment candidate. After Cell 4, <code>speed_mps</code> stored 5.2 m/s.

Pts.	Prompt	My response
1	Write one line you could run after Cell 2 to check the stored value of <code>speed_mps</code> without changing it.	Key: <code>print(speed_mps)</code> or bare <code>speed_mps</code> .
1	Write one line you could run after Cell 3 to check whether <code>adjusted_speed</code> exists.	Key: <code>print(adjusted_speed)</code> or bare <code>adjusted_speed</code> .
2	Before calculating speed, what measurement condition should the team check so the result is physically meaningful? Write it in plain language.	Key: Time must be positive/nonzero and correspond to the measured 12-meter path.

5. Compare two possible edits[debug](#)[implement](#)[explain](#)

The team considers two ways to apply the 0.2 m/s adjustment. Both can show 5.2; they do not leave the same evidence trail.

Version A: keep candidate separate

```
adjusted_speed = speed_mps + 0.2
print("stored", speed_mps)
print("adjusted", adjusted_speed)
```

Version B: overwrite immediately

```
speed_mps = speed_mps + 0.2
print("stored", speed_mps)
print("adjusted", speed_mps)
```

Pts.	Prompt	My response
1	In Version A, what is stored in <code>speed_mps</code> after the code runs?	Key: 5.0.
1	In Version A, what is stored in <code>adjusted_speed</code> after the code runs?	Key: 5.2.
1	In Version B, what is stored in <code>speed_mps</code> after the code runs?	Key: 5.2; immediate overwrite.
2	Which version gives a better audit trail for a lab record? Explain your choice.	Key: Version A, because measured speed and adjusted candidate remain separate until the record decision.
2	If the goal is to record the adjusted value only after review, write one sentence explaining why Cell 4 is a safer place to overwrite <code>speed_mps</code> .	Key: Cell 4 overwrites only after both measured and adjusted values have been made visible.

6. Mastery check and support next step**transfer****explain**

Use your own evidence from the workbooklet. Do not write a generic answer.

Pts.	Prompt	My response
2	What is one example from this notebook where the screen showed a value that was not yet the stored value that would be recorded?	Key: Cell 2 displayed 5.2, but <code>speed_mps</code> was still 5.0.
2	In one or two sentences, explain why engineers should preserve both visible output and stored-state evidence during early computational checks.	Key: Visible output shows what appeared on screen; stored-state evidence shows what the program will reuse or record. Keeping both prevents a displayed candidate from being mistaken for an approved/current result.

My mastery check

Mark the strongest statement that is true after this workbooklet.

☐	Secure: I can trace stored state and visible output separately, including a displayed value that is not stored yet.
☐	Developing: I can calculate some values, but I still need support separating display, print, assignment, and overwrite.
☐	Needs support: I need a smaller example before I can explain what Python stored after each cell.

My next support move

My issue is mostly: setup / Python syntax / tracing / arithmetic / notebook display / confidence / time.

The first place I got uncertain was: _____

My next small action is: ask a partner / ask instructor / rerun a cell / rebuild the trace table / try the recovery version.

Workbooklet target: I can build an evidence trail that separates measurement inputs, stored variable state, printed output, notebook display, and a later overwrite before a result enters an engineering record.

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Bridge Day 1 INSTRUCTOR KEY

VIP Evidence Handoff

Learning sequence: Predict - Trace - Explain - Test - Interpret

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Why engineers use this page

Carry exact-output and stored-value evidence directly into the first notebook, then complete its two-problem Independent Program Studio.

Decision boundary: Code is useful only when its stored state, visible result, assumptions, units, limits, and tests support a defensible engineering interpretation.

Construct readiness before independent work

New authoring tools: string literal, assignment, print call, sequential execution, exact output

Carried authoring tools: none

Supplied-code exposure only: none

An exposure-only construct may be traced in complete supplied code, but the independent task will not require students to invent it before its authoring day.

Notebook cells and task constructs

Activity	Work	Stable cells	Required authoring constructs
18.1	First Output	18-1-work / 18-1-transfer / 18-1-cold	profile-level contract
18.2	Formatted Notice	18-2-work / 18-2-transfer / 18-2-cold	profile-level contract

Readiness evidence

- For Activity 18.1, separate the exact fixed label from the value stored in the named variable.

Model response: The fixed label and stored value must be named separately; the final displayed line must match capitalization, colon, and spacing exactly.

- For Activity 18.2, number the required output lines and write the exact order before opening the notebook.

Model response: A complete response lists every line in the required order and treats spaces, punctuation, and capitalization as part of the specification.

Timing and help trigger

Record a first prediction before running. If you still lack an executable plan after about 8 focused minutes, use the linked ThinkCSPy refresher or the supplied model. If two attempts fail for the same named requirement, write the exact mismatch and ask a peer mentor or instructor a targeted question. Timing is diagnostic evidence, not a speed grade. **Start or elapsed minutes:** _____

My help trigger:

Engineering Evidence Record

Complete one record from a model, transfer, or Independent Program Studio build. Generic statements are not sufficient; use actual values, a real revision, and one concrete next test.

Evidence field	Record
Prediction	A specific expected state or output before execution.
Observed Result	The actual state or output, including a value or exact text.
Revision Reason	The defect or mismatch and the change that addresses it.
Engineering Meaning	How the evidence changes or supports the engineering decision.
Units Or Not Applicable	The physical unit, or the evidence type when no unit applies.
Assumption	One condition accepted as true for this model or rule.
Model Limit	One situation the result does not justify or predict.
Next Test	A concrete boundary, invalid, or alternate case with an expected result.
Help Trigger	The exact unresolved point that would trigger a targeted question.
Minutes Used	Approximate minutes from first prediction through revision.

Notebook and submission procedure

1. Attempt, predict, run, inspect, and revise before opening a reveal.
2. Complete the end-of-notebook Independent Program Studio without a reveal or copied solution. Only those visibly labeled Studio cells are scored for independent mastery.
3. Save or download the completed notebook as one `.ipynb` file.
4. Upload that one notebook to the matching Gradescope assignment. Do not export a `.py` file or create a ZIP.